



University of Asia Pacific
Department of Computer Science and Engineering
REPORT: 04

Course Code: CSE 402

Course Title: Operating System Lab

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Problem Statement:

Implement FCFS, SJF, and Round Robin CPU Scheduling for the following scenario, where the number of processes = 6 and the process information is:

P id+	AT	BT
P1	0	7
P2	1	4
P3	2	15
P4	3	11
P5	4	20
p6	4	9

Calculate CT, TAT, WT, Average TAT and Average WT for FCFS, SJF and Round Robin. For Round Robin, use **Time Quantum = 5**. Also represent the process execution sequence and compare the performance of the three scheduling algorithms.

Why is Round Robin Used in CPU Scheduling?

Round Robin is mainly used in time-sharing and multitasking systems. Each process gets a fixed amount of CPU time called the Time Quantum. If a process does not finish within that time, it goes back to the ready queue.

This makes Round Robin more fair because one process cannot keep the CPU for a very long time. It is especially useful when many processes need to share the CPU.

Result OF FCFS:

PID	AT	BT	CT	TAT	WT
P1	0	7	7	7	0
P2	1	4	11	10	6
P3	2	15	26	24	9
P4	3	11	37	34	23
P5	4	20	57	53	33
P6	4	9	66	62	53

Average TAT = 31.67

Average WT = 20.67

Result OF SJF:

PID	AT	BT	CT	TAT	WT
P1	0	7	7	7	0
P2	1	4	11	10	6
P3	2	15	46	44	29
P4	3	11	31	28	17
P5	4	20	66	62	42
P6	4	9	20	16	7

Average TAT = 27.83

Average WT = 16.83

Result OF RR:

PID	AT	BT	CT	TAT	WT
P1	0	7	31	31	24
P2	1	4	9	8	4
P3	2	15	55	53	38
P4	3	11	56	53	42
P5	4	20	66	62	42
P6	4	9	50	46	37

Average TAT = 42.17

Average WT = 31.17

Execution Sequence:

FCFS:

P1 → P2 → P3 → P4 → P5 → P6

SJF:

P1 → P2 → P6 → P4 → P3 → P5

Round Robin (TQ = 5):

**P1 → P2 → P3 → P4 → P5 → P6 → P1 → P3 → P4 → P5
→ P6 → P3 → P4 → P5 → P5**

Algorithm	Avg WT	Avg TAT
SJF	16.83	27.83
FCFS	20.67	31.67
RR	31.17	42.17

For the given processes, SJF performs better than FCFS and Round Robin because it provides the lowest average waiting time and turnaround time. However, Round Robin is more suitable for time-sharing systems because it gives each process a fair share of CPU time.

Discussion:

In this experiment, three different CPU scheduling algorithms were implemented and compared. FCFS is simple because it follows the arrival order, while SJF gives priority to processes with shorter burst times. Round Robin gives every process a fixed CPU time, making it more suitable for multitasking systems. For the given process set, SJF produced the lowest average waiting time (16.83), while FCFS had 20.67 and Round Robin had 31.17. Therefore, SJF performed better in terms of average waiting time for this particular input.

Conclusion:

The experiment successfully implemented FCFS, SJF and Round Robin scheduling using Python. The results show that the performance of a scheduling algorithm depends on the process characteristics. For this particular scenario, SJF provided the lowest average waiting time, while Round Robin provided fair CPU sharing through a fixed time quantum. This experiment helped demonstrate the practical differences between the three scheduling techniques.

Git Repository link :

<https://github.com/Rajash144/OS-CPU-Scheduling-Labs>